

391 Computer Programming Concepts (Open)

(formerly 390)

Description & Eligibility

Contestants will demonstrate foundational knowledge of computer programming concepts, including languages, logic, algorithm design, data structures, and core software development principles. Any secondary or postsecondary division contestant may enter this objective event.

Contestant Must Supply

- (Optional) Cordless calculator: Electronic devices will be monitored according to ACT standards. See NLC Calculator Guidelines. Contestants who violate this rule will be disqualified

Contest Notes

- Method of evaluation: objective
- AI Policy Level 1 *No AI Allowed*
- Reference materials are not allowed. Contestants who violate this rule will be disqualified
- Length of event: no more than sixty (60) minutes testing time
- Unlimited entries by state

Contest Competencies

1. Demonstrate foundational understanding of computer science principles, including logic, data structures, and algorithms, and demonstrate proficiency in modern programming languages with adaptability to others as needed
2. Design and implement efficient algorithms using structured and object-oriented design techniques to create scalable and maintainable software solutions
3. Develop applications using control flow, logical operations, strings, arrays, classes, file I/O, loops, and advanced data structures to process and manage data effectively
4. Apply the software development lifecycle and create design artifacts such as UML diagrams, flowcharts, and pseudocode to plan, implement, and communicate system architecture
5. Write and optimize SQL queries to interact with relational databases for data storage, retrieval, and analysis

ARTS, ENTERTAINMENT, & DESIGN

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- 405 Advanced Desktop Publishing
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- 420 Digital Media Production
- 425 3D Computer Modeling
- 430 Video Production Team
- 440 3D Computer Animation Team
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